Open Name Tags (ONT)

A short, human-readable name — like `alice` — that you truly own. design brief · v0 prototype

Most names online are really *accounts*: a company hands them out, and can rename you, reclaim them, or shut you down. An ONT name is different — it is controlled by a cryptographic key that only you hold. No company is involved, there is nothing to renew or pay rent on, there is no token, and no one (not even ONT's creators) can move or revoke it. Anyone can look up who owns a name and confirm the answer for themselves.

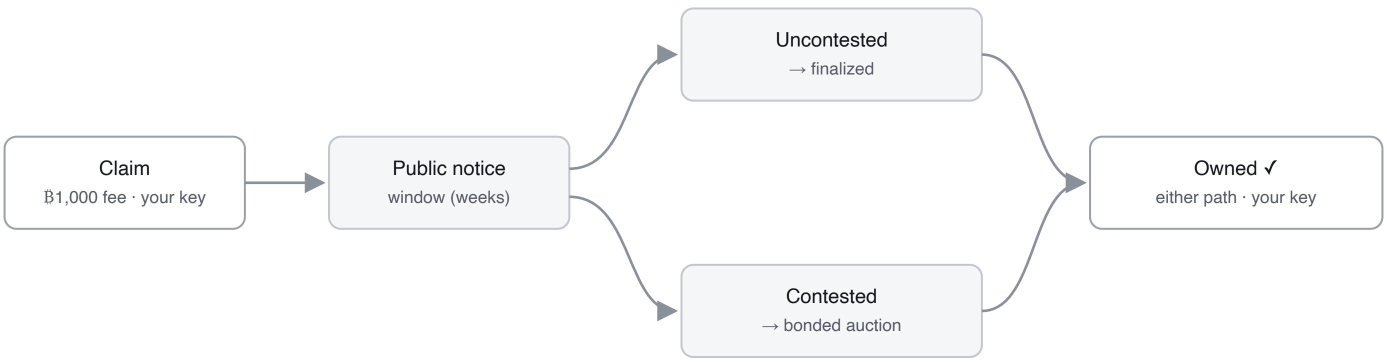
WHAT YOU'D USE ONE FOR

- **A payment handle** — pay `alice` instead of a long address; the name points to wherever she wants to be paid.
- **An identity handle** — one username for open-source / decentralized apps that no platform can reassign or revoke.

HOW YOU GET A NAME

There is one path for every name. It forks only if two people want the same one.

1. **Claim it.** Pay a small, one-time Bitcoin fee — about $1 (**Ƀ1,000**) — paid to Bitcoin's miners, not to ONT.
2. **A public notice window opens** (a few weeks). This is everyone else's chance to contest the name — if someone else wants it too, that sends it to the auction in step 4 instead of finalizing cheaply.
3. **If no one else does, the name is yours.** Thousands of these uncontested claims are bundled into one small Bitcoin record — which is what keeps it cheap across billions of names.
4. **If someone else wants it too, the name is *contested*** and goes to auction: each bidder locks up bitcoin as a **returnable ~one-year bond** (they keep custody; it's released at maturity), and the largest bond wins. Contests can be common early (everyone wants `bitcoin`, popular brands, or dictionary words) but are rare across the long tail at scale, so for most names the auction never happens.



Either path ends in the same place — you own the name, controlled by your key. Most names finalize at window close; a contested one is settled by auction (largest returnable bond wins). You can use it the moment the auction **settles** — the bond just stays posted through maturity (~1 yr) then returns, an ONT rule (break it early and you forfeit the name), not a Bitcoin timelock.

WHAT OWNING A NAME LETS YOU DO

A name is controlled by one key — your **owner key**. With it you can:

- **Point the name somewhere** — at a Bitcoin or Lightning address, a website, and so on — and change it whenever you like. (These mappings live off-chain, signed by your key, so updates are instant, free, and never touch Bitcoin.)
- **Transfer** the name to someone else's key.
- **Optionally set up recovery** so a lost key isn't the end — opt-in (skip it and a name is just one key you keep safe, like cold-storage bitcoin). Only your chosen backup key can use it, and your main key vetoes misuse within a window — a veto you can hand to a non-custodial watcher, so a set-and-forget name needn't keep you online.

TWO SERVICES THAT HELP — NEITHER DECIDES

Using ONT at scale leans on two **unprivileged** services; neither decides ownership — Bitcoin does. A **publisher** is the service you *pay* (over Lightning) to get your claim into Bitcoin — it batches thousands of claims and broadcasts the on-chain anchor (needs a Lightning rail + on-chain funds). A **resolver** is the service you *query* to look up a name — it replays Bitcoin and serves the answer, never deciding it (needs only a Bitcoin node + storage). One operator usually runs both, but they're **separate at the protocol layer**: claim through one, verify against another, or self-host either.

HOW IT SCALES

Billions of names can't each be a Bitcoin transaction, so **publishers** batch many claims into one Merkle commitment and anchor only its root — a ~150-byte root that commits to the whole batch *whatever its size*, so the more you batch the lower the per-name cost (~**0.015 vB**/name at ten thousand per batch, less as batches grow). A batch counts only if its miner fee covers the claims inside it, so each name still buys the blockspace it uses.

You pay a publisher off-chain over Lightning; it bundles many claims and pays the single aggregate miner fee — so the ₿1,000 still reaches **Bitcoin's miners**, even though your immediate payment goes to the publisher (it isn't a fee to the publisher or a resolver). **Your cost is the ₿1,000 gate (sunk, to miners) plus a thin publisher service fee** — the publisher's own per-name cost is tiny, and any markup is capped by the always-available option of claiming directly on L1. The flow is **pay-first** (you pay, then you're included; a non-payer is left out), so the publisher risks no capital — you take a small, bounded one. And a publisher **can't steal a name**: if it pockets your payment or commits the wrong owner key, you contest on-chain, which forces an auction the rightful owner wins; worst case you're out about a dollar and re-claim elsewhere. Binding the payment to inclusion atomically is a possible future refinement, not a v1 dependency. **v1 starts with a few reputable publishers and minimizes even that small trust over time.**

WHY YOU CAN TRUST IT

No company, server, or founder decides who owns a name — Bitcoin does. The rules that turn Bitcoin transactions into ownership live in a small, **frozen core — three consensus files** that anyone can audit, locked so its trust surface can't silently grow. Run it over Bitcoin's history and you get the same answer everyone else does, and you can check that answer against Bitcoin's own block headers and proof-of-work — so a server that lies about who owns a name gets caught, not believed. Resolvers only mirror this Bitcoin-derived data; they never decide it.

And no operator is privileged: **anyone can run a resolver or publisher** (the two roles above). Because ownership is fixed by Bitcoin, you don't need a *trusted* node — only a reachable one whose answer you can verify (a lying node is caught; a slow one is routed around). Finding nodes is config-seeded today; a registry-free, on-chain discovery scan is designed, not yet built.

THE NUMBERS WE'RE PROPOSING — SEVERAL ARE PLACEHOLDERS, ALL OPEN TO CHALLENGE

PARAMETER	PROPOSED	STATUS
Claim fee (every name)	**₿1,000** ~$1, sunk, to miners	baseline
Contested-auction min bond	**₿50,000** ~$50, returnable	placeholder
Bond maturity	~52,560 blocks (~1 yr)	test override
Notice window	weeks, height-keyed	placeholder · fairness lever
Data-availability windows	unset	deadline for batch bytes to surface + reorg depth; unpinned
On-chain footprint	~0.015 vB/name; fee = Σ fees	measured

Opening bond for scarce short names.

NAME LENGTH	OPENING BOND	~USD
1 char	**₿100,000,000** (1 BTC)	~$100k
2 char	₿50,000,000	~$50k
3 char	₿25,000,000	~$25k
4 char	₿12,500,000	~$12.5k
5+ char	fee only; ₿50,000 floor if contested	~$1 / ~$50

Only the very short (≤4-char) names carry a high length-scaled opening bond (halving per added character); everything else just uses the flat fee plus a bond if contested.

Least sure of: the **contest rate** is unknown until launch — we assume it's high early (everyone wants `bitcoin`, popular brands, or dictionary words) and low for the long tail (`sallysmith2165`); and the notice window has to be long enough for a competitive early market to form, so premium names aren't swept cheaply before other bidders show up.

STATUS — HONEST (MATURITY, NOT DIRECTION)

Live on a Bitcoin test network (signet), end-to-end: claim, owner-key transfer, owner-signed records, recovery, and a bonded auction bid the resolver accepts — with the consensus code and signatures cross-checked byte-for-byte against a second independent implementation. **Prototype / partial:** the cheap batched-claim path is now wired into the live indexer — it observes the anchored root chain and resolves accumulator-claimed names, re-verifying each against Bitcoin — but the batch-data *transport* is still a pluggable seam (not a production source) and the resolver/web surface doesn't expose those names yet; a single-writer publisher; and producers don't yet emit the proofs a phone/browser would check. Not mainnet-ready.

WHAT WE MOST WANT YOU TO PUSH ON

- **Data availability — incl. how the bytes are transported** — we batch claims and anchor only a summary on Bitcoin, leaving the claim data off-chain; our defense if someone withholds it (or a reorg reshuffles it) is a deadline — data that isn't public by a set Bitcoin height simply doesn't count. Is that sound, and should availability be proven on-chain or by timing alone? And the *transport*: we lean toward content-addressed bytes any node can mirror, verified against an on-chain digest (so it's **not consensus-critical**, backend swappable) — is publisher-served + voluntary mirrors enough for v1, or is a gossip/DA-sampling layer needed sooner?
- **Publisher trust-minimization** — v1 leans on reputable, pay-first publishers (a non-payer is left out). Is there a clean, *deployable-today* way to bind "pay the publisher" to "claim anchored on-chain" without depending on long-roadmap primitives — or is reputable-publisher trust the right v1 stance, with atomic binding left as later research?
- **Discovery & censorship-resistance** — config-seeded today; is a registry-free, on-chain service-announcement scan the right trustless discovery primitive, with Bitcoin + verification as the only trust root?
- **On-chain footprint** — are the ~171-byte OP_RETURN events acceptable on mainnet, or is a script/covenant carrier worth a soft-fork dependency?
- **Light-client verification** — a launch blocker, or fine post-launch?
- **Auction form** — open ascending vs. sealed second-price, given MEV and relay-bid timing?
- **Launch fairness** — is a long notice window enough against a day-one rush on premium names, or do we need a decaying launch fee?
- **Recovery you can leave unattended** — recovery is opt-in (skip it and a name is one key, like cold-storage bitcoin); if enabled, its veto should be postable by a *non-custodial watcher* (can abort, never move the name), not the owner. A literal pre-signed veto can't reference the recovery UTXO (outpoint unknown until invoke) — what's the cleanest name-scoped, abort-only credential?

Repo: github.com/deekay/ont · the full design brief, prior-art comparison, risk register, and the verification you can run yourself: `docs/ONT_DESIGN_BRIEF.md`.